

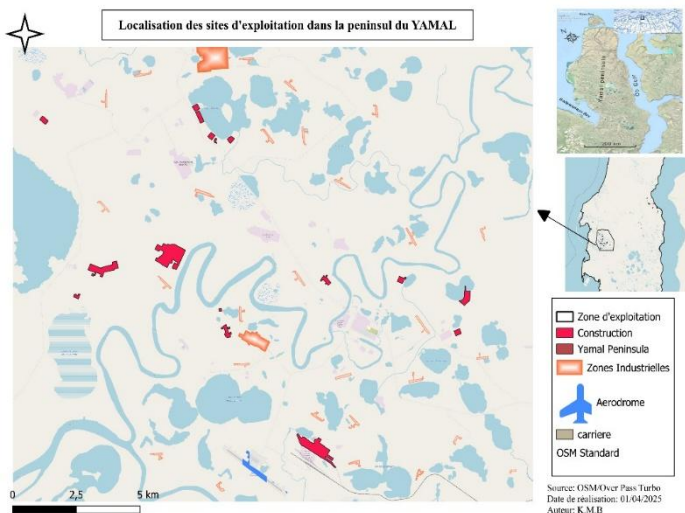
YAMAL PENINSULA OIL AND GAS DEVELOPMENT

Impacts of operations: atmospheric pollution and degradation of vegetation

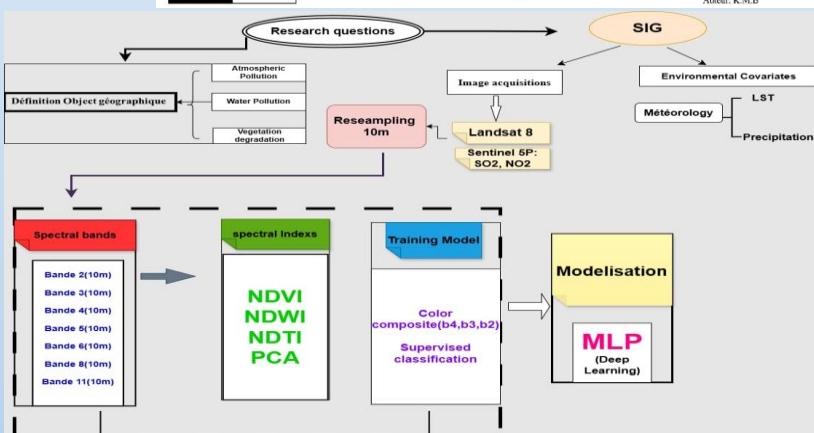
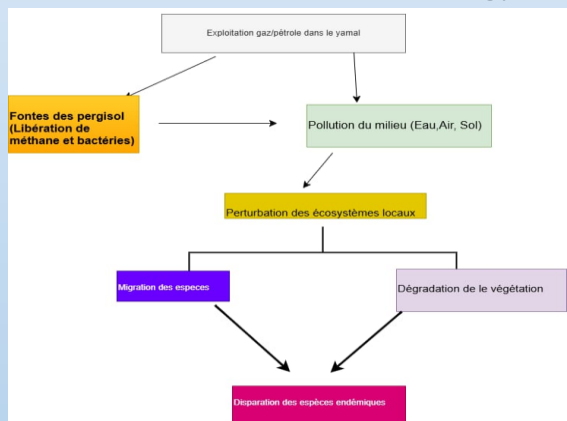
Introduction

The Yamal Peninsula, located in Western Siberia, is a region rich in natural resources, particularly gas and oil. For several decades, this area has been subject to intensive exploitation by the energy industry, drawn by its vast deposits. However, this economic activity is not without consequences for the fragile Arctic environment. Indeed, the extraction of gas and oil causes major impacts such as air pollution, linked to industrial emissions, as well as the gradual degradation of vegetation, which is essential to maintaining the balance of local ecosystems.

Location of the study area

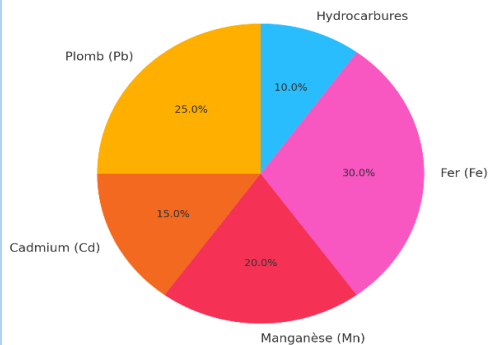


1. Methodology



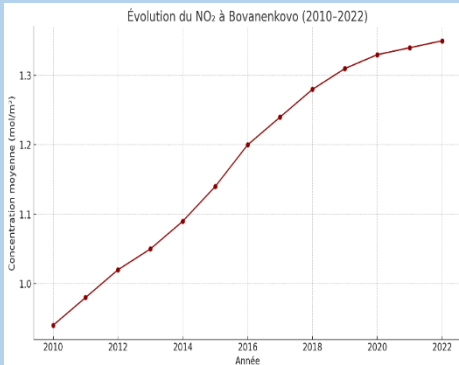
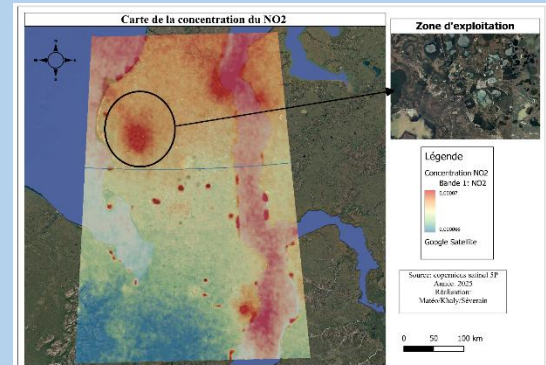
2. Water pollution on the Yamal peninsula

Répartition estimée des polluants dans les eaux de surface



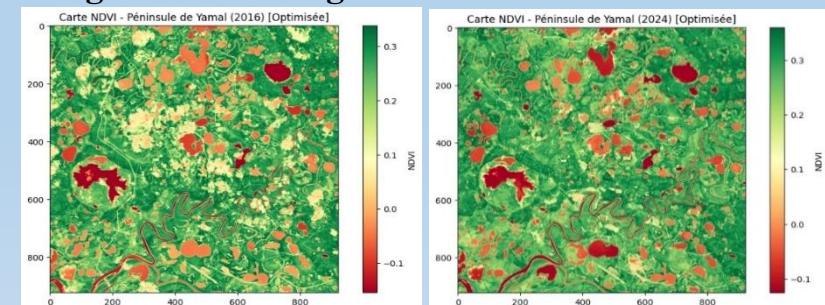
Polluant	Probable source	Risks
Plomb (Pb)	Industrial waste, pipe corrosion	Neurotoxicity, hypertension
Cadmium (Cd)	Fossil fuels, gas pipelines	Carcinogenic, kidney disorders
Manganèse (Mn)	Natural soils + industrial flaring	Neurological disorders
Fer (Fe)	Natural overload + pollution	Metallic taste, deposit
Hydrocarbures	Accidental or chronic discharges	Biological disturbance, bioaccumulation

3. NO2 concentration in the atmosphere



- ❑ Average NO₂ concentration around Bovanenkovo: 1.34 mol/m³
- ❑ +43% average increase in NO₂ concentrations around industrial sites between 2010 and 2022.
- ❑ TROPOMI images show an expansion of affected areas within a radius of 15 to 20 km around Bovanenkovo.

4. Degradation of vegetation



- ❑ **Active vegetation:** present on both maps, but in 2024 a more heterogeneous distribution can be observed; some green areas have become fragmented and others have slightly decreased.
- ❑ **Landscape modification:** an increase in water bodies and wetland areas at the expense of stable vegetated surfaces.

Conclusion

In conclusion, oil and gas exploitation in the Yamal Peninsula has significant environmental consequences, such as air pollution and vegetation degradation, which threaten the region's fragile ecosystems. In the future, the use of Deep Learning models could help to better monitor these impacts and suggest solutions to reduce environmental damage.